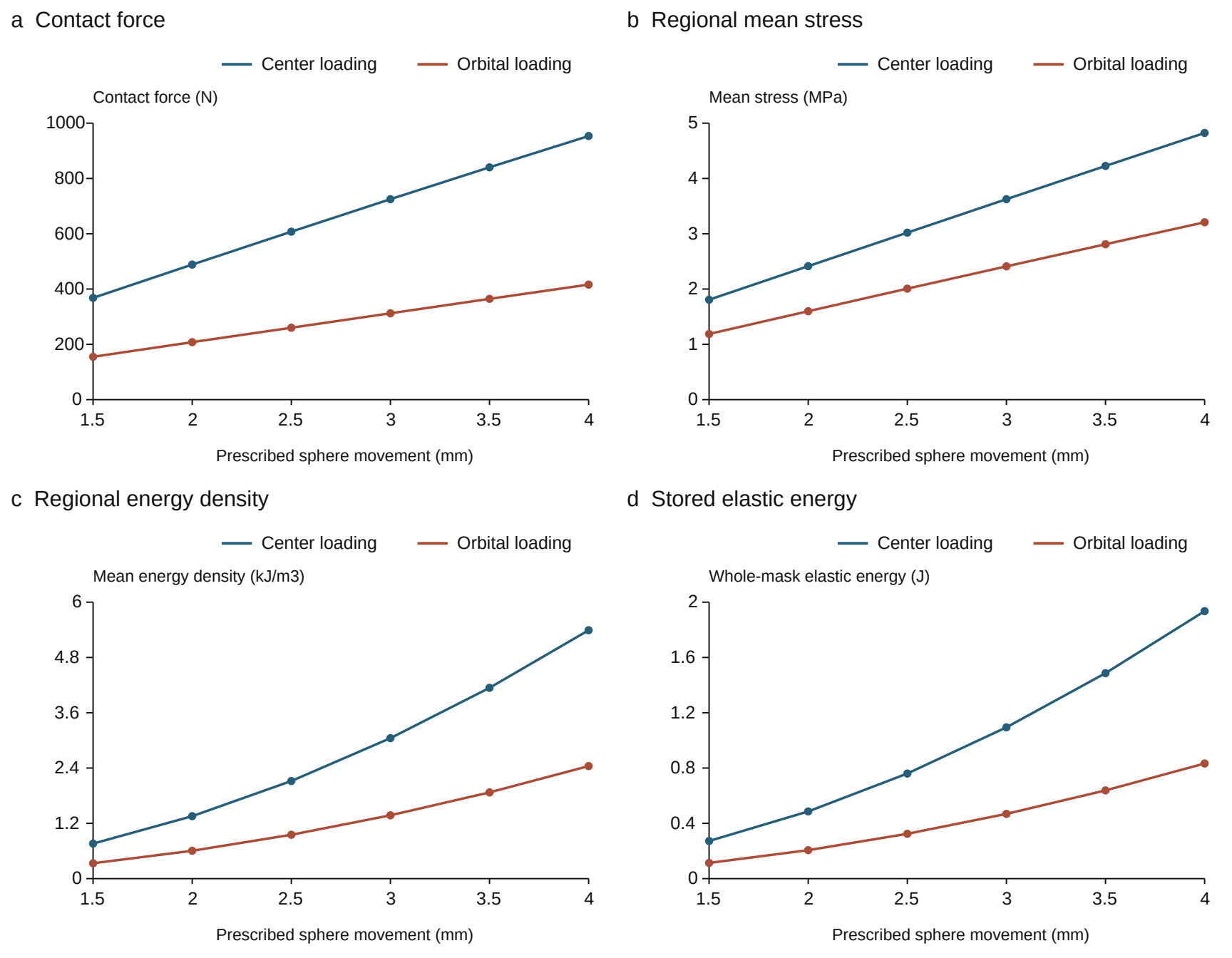
a Contact force
Center loading
Orbital loading
Contact force (N)
1000
800
600
400
200
0
1.5
2
2.5
3
3.5
4
Prescribed sphere movement (mm)
b Regional mean stress
Center loading
Orbital loading
Mean stress (MPa)
5
4
3
2
1
0
1.5
2
2.5
3
3.5
4
Prescribed sphere movement (mm)
c Regional energy density
Center loading
Orbital loading
Mean energy density (kJ/m3)
6
4.8
3.6
2.4
1.2
0
1.5
2
2.5
3
3.5
4
Prescribed sphere movement (mm)
d Stored elastic energy
Center loading
Orbital loading
Whole-mask elastic energy (J)
2
1.6
1.2
0.8
0.4
0
1.5
2
2.5
3
3.5
4
Prescribed sphere movement (mm)